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Section: F

Chat History Link: <https://claude.ai/share/bd9dd055-d113-4ad1-ba6b-66fb48cb4b3d>

game_engine.py
150 lines
PY

marble.py
13 lines
PY

wall.py
12 lines
PY

main.py
108 lines
PY

Task 1: Refine Collision Detection
Near a wall's corner, the marble sometimes bounces off empty space before it visually touches the wall. Investigate and enhance collision accuracy so bounces only happen when the round marble actually touches a wall.

Task 2: Implement Game Over Condition
Add a screen that displays whether the maze was solved (with the finish time) or the timer ran out, then gracefully waits for input instead of just printing to the console.

Task 3: Add Replay Option
After the end screen, allow the user to play again by choosing a difficulty (Easy, Medium, or Hard tilt strength/friction/time limit), or exit.

Task 4: Add Sound Feedback
Add basic sound effects for bouncing off a wall, reaching the goal, and the timer running out.

Expected Behavior

- The marble accelerates toward the mouse cursor's direction relative to the screen center, simulating a physical tilt
- Friction gradually slows the marble down, and its top speed is capped
- The marble bounces off walls instead of passing through them
- Reaching the goal circle ends the round as a win; running out of time ends it as a loss
- A countdown timer is visible at all times

Please provide the fixed version of all the files w.r.t the tasks and expected behaviour. Don't make any mistakes

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3 hours ago ↻ ↗ 📄

Ran 5 commands, created 5 files, and 2 more tools >

I've fixed all four tasks and tested them headless (pygame 2.6.1, dummy video and audio drivers). I haven't played it with a real mouse and speakers, and in the test the mouse sat at (0,0), so that run only checked that the marble stays inside the window.

The layout is `main.py` plus a `game/` folder (`game_engine.py`, `marble.py`, `wall.py`, and a new `sounds.py`). The folder needs an `__init__.py`, which I added empty for the test.

Task 1: collision. The old code compared the marble's square bounding box with the wall, so a corner could trigger a bounce while the round marble was still clear of it. It now finds the point on the wall closest to the marble's centre and bounces only if that point is within the radius. I tested a marble sitting at a wall corner, where its bounding box overlaps the wall but the circle doesn't. The old code would have bounced it, and the new one correctly doesn't.

- The marble is pushed out along the contact direction, and only the speed into the wall is reversed (0.3, same as before), so it slides along flat walls and bounces off a corner at the right angle.
- If the centre ends up inside a wall, it is pushed out through the nearest face.
- `Wall` gained a `closest_point` helper, and `Marble` gained a `reset` method.

Task 2: end screen. Instead of printing to the console, a panel appears over the frozen maze. It shows "Maze Solved!" with the finish time, or "Time's Up!". The timer freezes when the round ends, and the countdown now rounds up so it doesn't show 0s while time remains. Input is ignored for 0.5s so a stray keypress doesn't skip the screen.

Task 3: replay. The same panel offers Easy, Medium and Hard buttons, plus Exit. You can click them or press 1/2/3 (or E/M/H), and Esc or Q exits. The first game uses Medium, which matches your original values.

	Tilt strength	Friction	Time limit
Easy	0.5	0.04	60s
Medium	0.6	0.02	45s
Hard	0.8	0.008	30s

I picked the Easy and Hard values myself, so change them in `DIFFICULTIES` in `game_engine.py` if you want a different feel.

Task 4: sound. The three effects are generated in code, so there are no audio files to ship.

- **Bounce:** a low thud, louder for harder hits. It only plays on real impacts, so a marble resting against a wall stays quiet.
- **Goal:** a rising four-note jingle.
- **Timeout:** a falling buzzer.

If there is no audio device, the game runs silently instead of crashing. `main.py` only changed in that it sets up the audio mixer before `pygame.init()`.

 Game engine
PY

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wait can you merge sounds with other file?

Merging sound functionality into the main engine file.

Sure. I'll move `SoundManager` into `game_engine.py`, delete `sounds.py`, and rerun the tests.

Ran 2 commands, shared files >

I merged `SoundManager` into `game_engine.py`, so it now sits just above the `GameEngine` class. I deleted `sounds.py`, and the only import change is the extra `import array` at the top. The headless tests pass again after the merge.

You now need just these files: `main.py` and `game/` with `__init__.py`, `game_engine.py`, `marble.py` and `wall.py`. `marble.py`, `wall.py` and `main.py` are unchanged from my previous reply, so only `game_engine.py` needs replacing.

One small thing: a leftover `__pycache__` folder in my output directory wouldn't delete because of a file-system error. It's just Python's cache and is safe to ignore or delete.

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